Pre-class progress check

By now, you should have

1. Reviewed slides from weeks 1 — 4
2. Reviewed code from classes 1 — 4
3. Attempted quizzes 1 - 4

1. Checked assignment 1 (due 10.12)
2. Attempted Quiz 5 (released on 09.29)
3. Started thinking about final project proposal (due on 10.12 - tentative)

1. Checked out slides for today

University of Pittsburgh | School of Public Health

Supervised learning (II): Classification (II)
- regularization for classification
- support vector machines

BIOST 2155

Class 6
October 3rd, 2025

fig1

Support vectors

Introducing hyperplanes

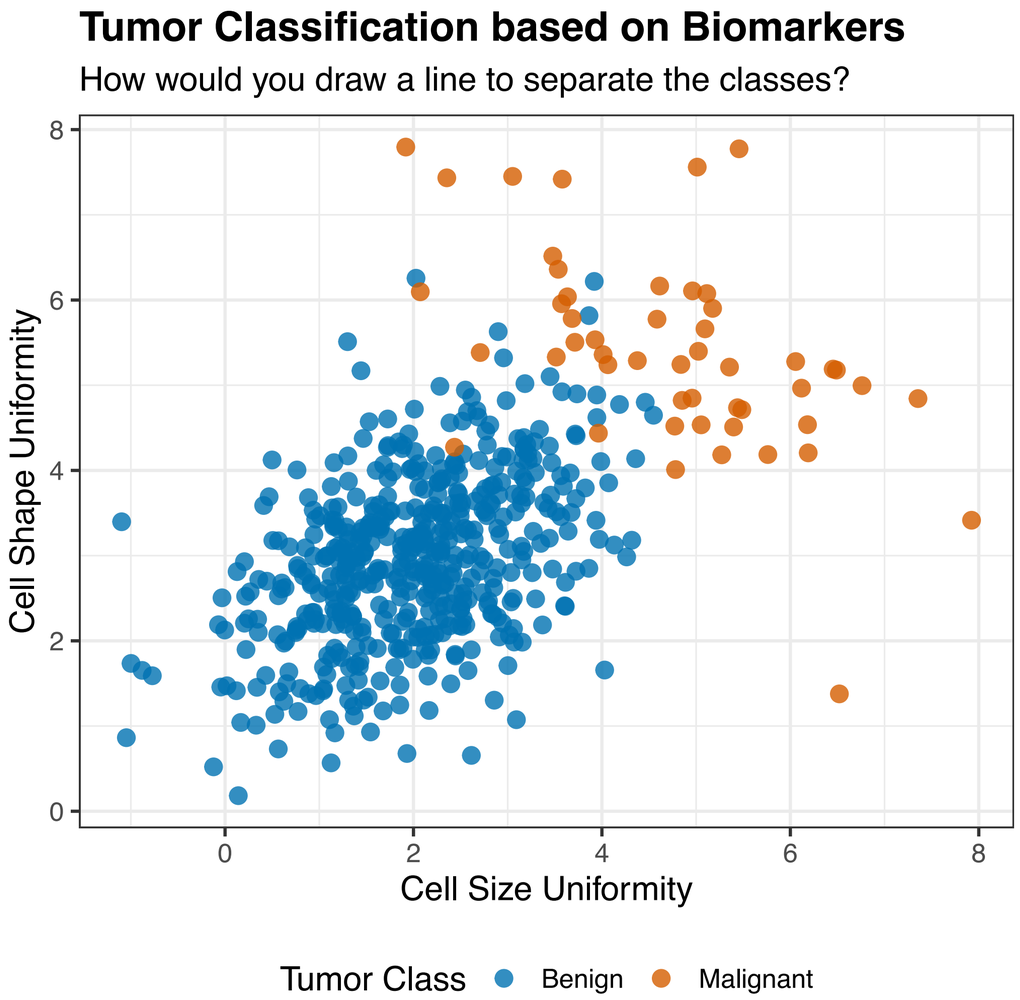
Key idea: We try and find a plane that separates the classes in feature space

If we cannot, we get creative in two ways:

- We soften what we mean by "separates", and
- We enrich and enlarge the feature space so that separation is possible.

Consider the following motivating example:
Want to classify patient tumors as benign or malignant based on biomarkers.

Two features: `Cell Size Uniformity` and `Cell Shape Uniformity`.



We will focus on binary classification for now and extend to multi-class problems later

What is a hyperplane?

Some math + visualizing for $p = 2$ dimensions

- Hyperplane in p dimensions is a <u>flat affine subspace</u> of dimension $p - 1$
- General the equation for a hyperplane has the form

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = 0$$

- For $p = 2$ we have a straight line

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

- The vector $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_p)^T$ is called the **normal vector** - it points in a direction orthogonal to the surface of the hyperplane.

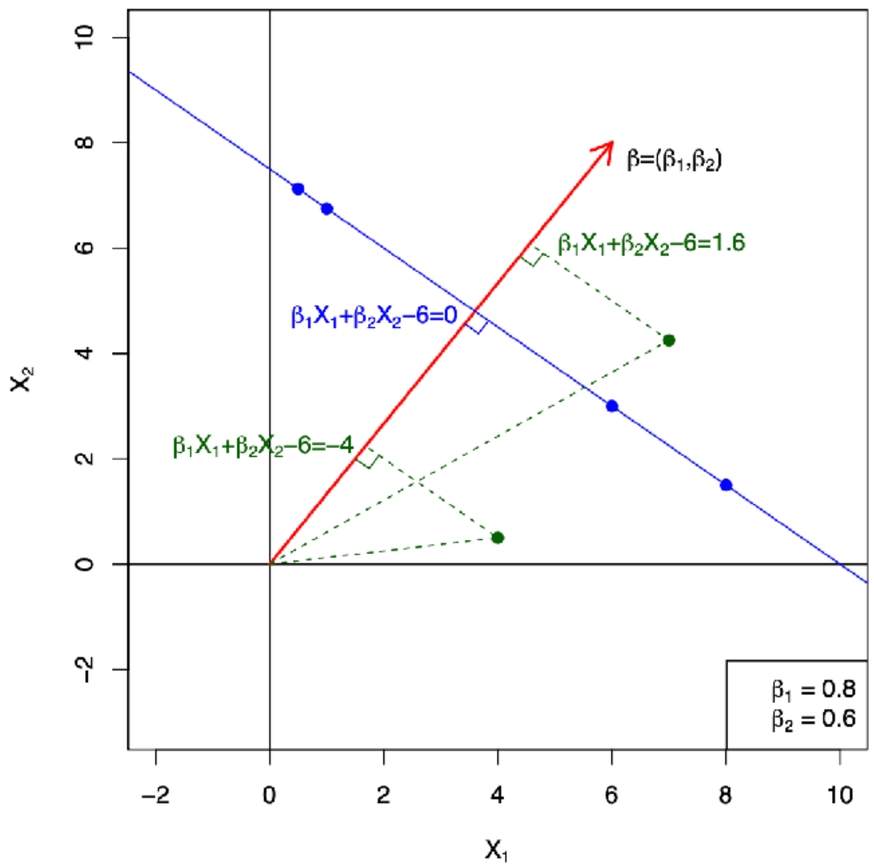


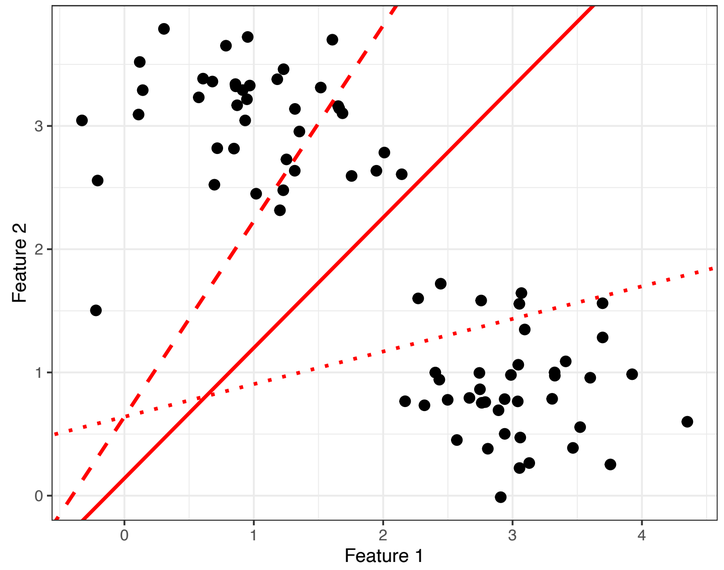
fig2

Separating hyperplanes

$f(\boldsymbol{X})$ **describes a hyperplane + visualizing for** $p = 2$ **dimensions**

$$f(\boldsymbol{X}) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

- $f(\boldsymbol{X}) > 0$ for points on one side of the hyperplane
- $f(\boldsymbol{X}) < 0$ for points on other side.
- Given features $\boldsymbol{X} = (X_1, \ldots, X_p)^T$, assign **<u>numeric</u>** label ($Y = \pm 1$) based on $f(\boldsymbol{X}) \lessgtr 0$
- If $Y_i \cdot f(\boldsymbol{X}_i) > 0$ for all i, then $f(\boldsymbol{X}) = 0$ describes a separating hyperplane.



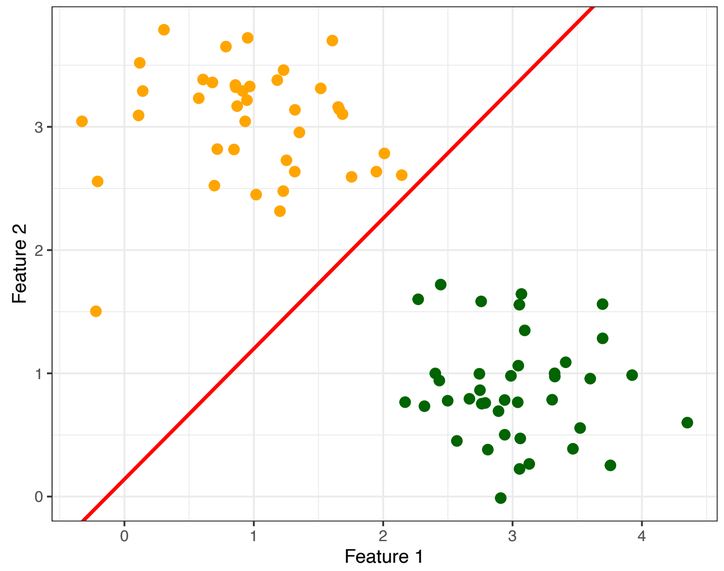


fig2

Maximal margin classifier (MMC)

MMC = hyperplane that provides strongest separation

Among all separating hyperplanes, find the one that makes the biggest gap or margin between the two classes.

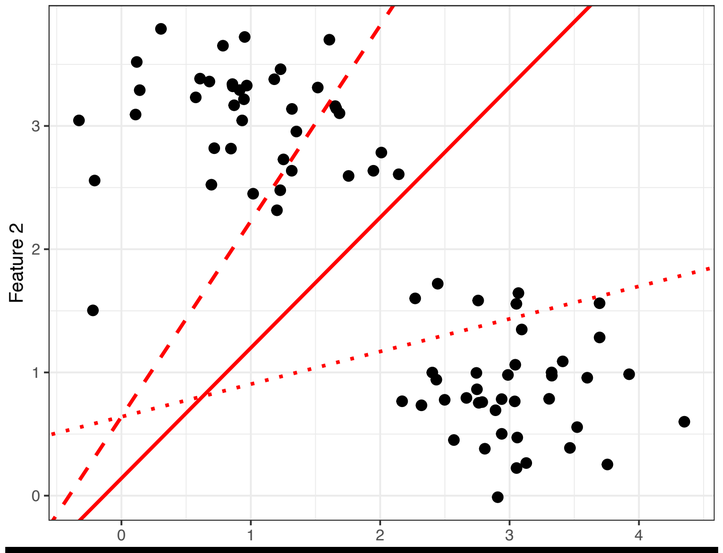
Maximize M with respect to $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$ subject to

1. $\sum_{j=1}^{p} \beta_j^2 = 1$
2. $y_i(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}) \geq M$ for all $i = 1, \ldots, n$.

Constraint #2 defines a (candidate) hyperplane.

Constraint #1: distance from the i-th observation to the hyperplane is

$$y_i(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip})$$



M = distance between red and blue lines

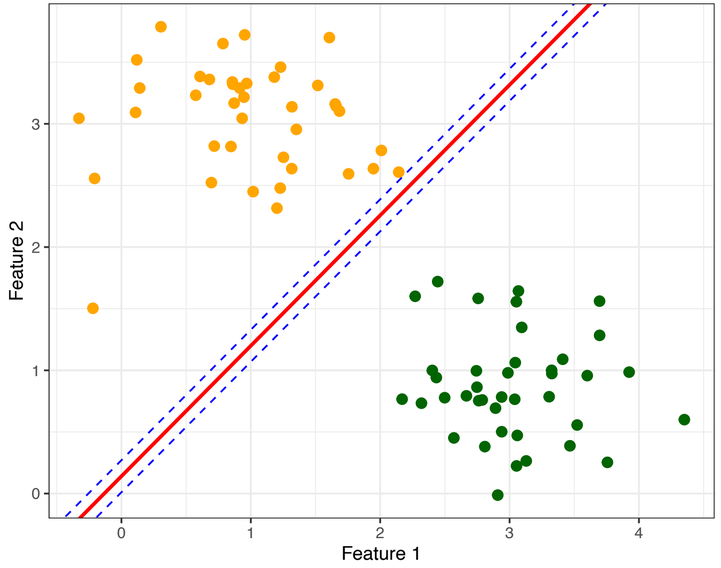


fig2

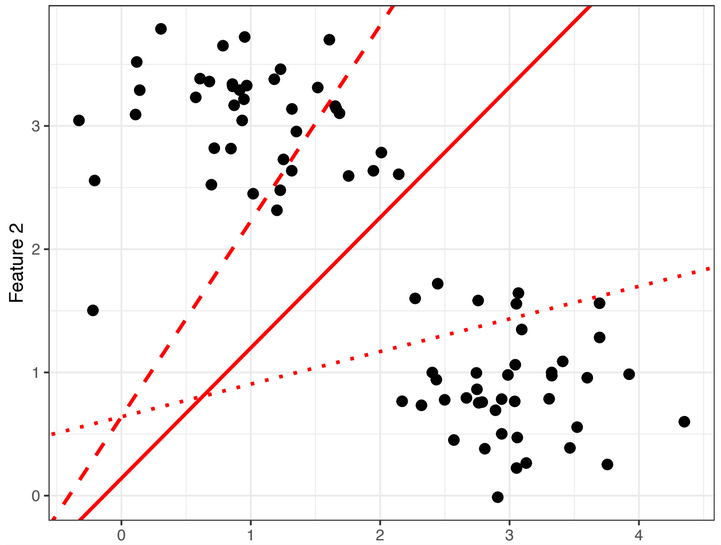
Support Vectors

Points on the edge of (dashed) margins

The observations that lie on the edge of the margin are called support vectors. These are the most critical data points; if you move them, the hyperplane may move.

Points further from the margin have no influence on the classifier.

Key Limitation: The MMC only works if the data is perfectly linearly separable. What happens in the real world when classes overlap?



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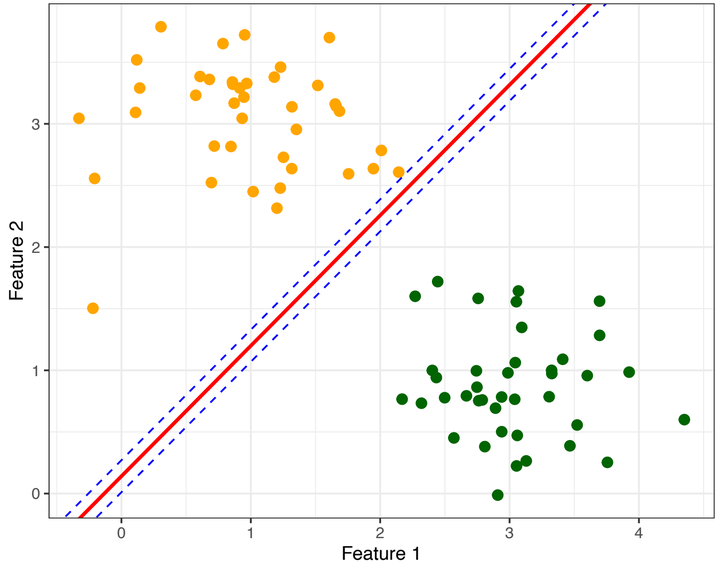


Support Vectors in the wild

Linear Support Vector Classifier (Soft Margin)

What if the classes overlap? No separating hyperplane exists!

We can allow some observations to be on the wrong side of the margin (or even the hyperplane). This is a "soft margin". Introduces a trade-off:

1. Maximize the margin width.
2. Limit the number of "margin violations"

Maximize M with respect to $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$ and $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_n)^T$ subject to

1. $\sum_{j=1}^{p} \beta_j^2 = 1$
2. $y_i(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}) \geq M(1 - \epsilon_i)$ for all $i = 1, \ldots, n$.
3. $\epsilon_i \geq 0$ and $\sum_{i=1}^{n} \epsilon_i \leq C$

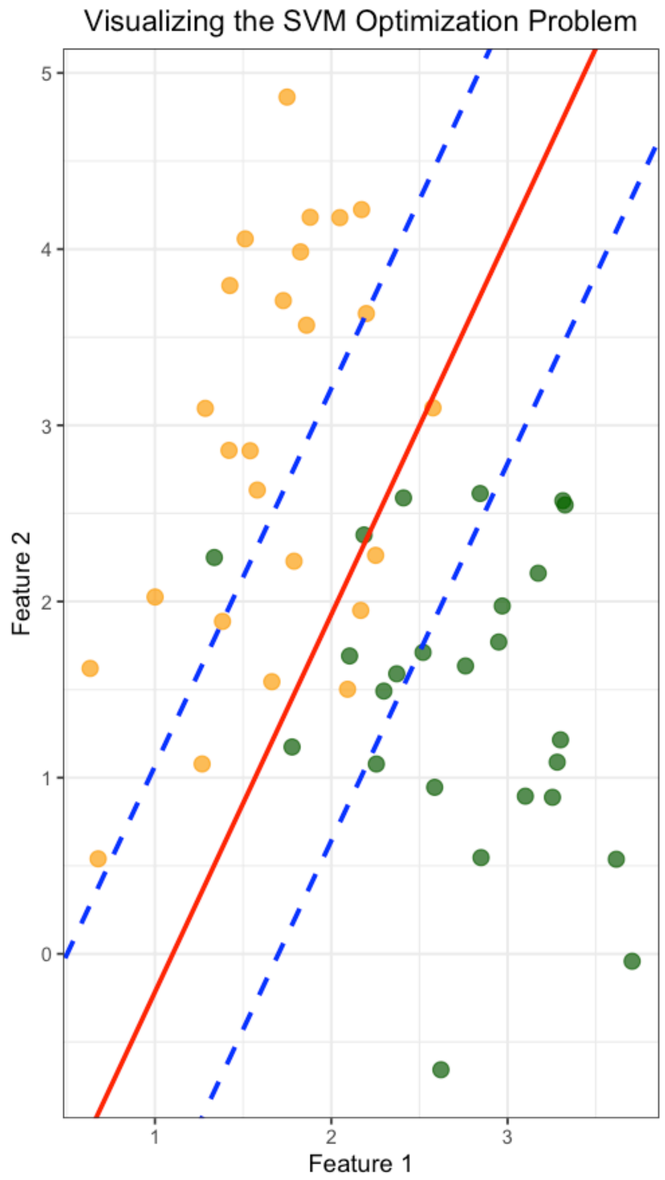
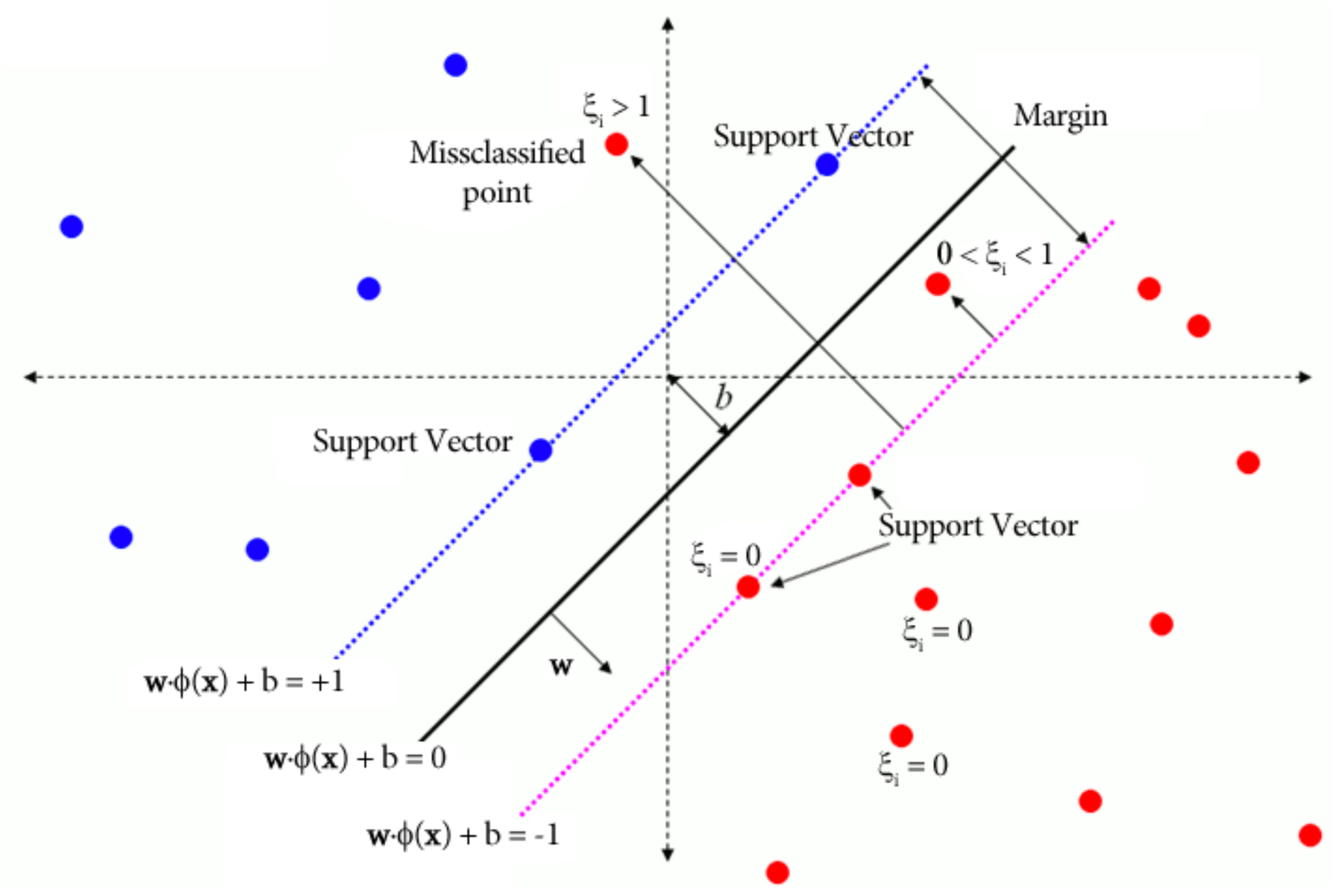


fig3

ϵ tells us where a point is located.

$\epsilon = 0$: point is on correct side of margin
$\epsilon > 0$: wrong side of margin
$\epsilon > 1$: wrong side of hyperplane


$\xi_i > 1$
Missclassified point
Support Vector
Margin
$0 < \xi_i < 1$
b
Support Vector
Support Vector
$\xi_i = 0$
$\xi_i = 0$
$\xi_i = 0$
$\mathbf{w}$
$\mathbf{w}\cdot\phi(\mathbf{x}) + b = +1$
$\mathbf{w}\cdot\phi(\mathbf{x}) + b = 0$
$\mathbf{w}\cdot\phi(\mathbf{x}) + b = -1$

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3. $\epsilon_i \geq 0$ and $\sum_{i=1}^{n} \epsilon_i \leq C$

How to choose C?

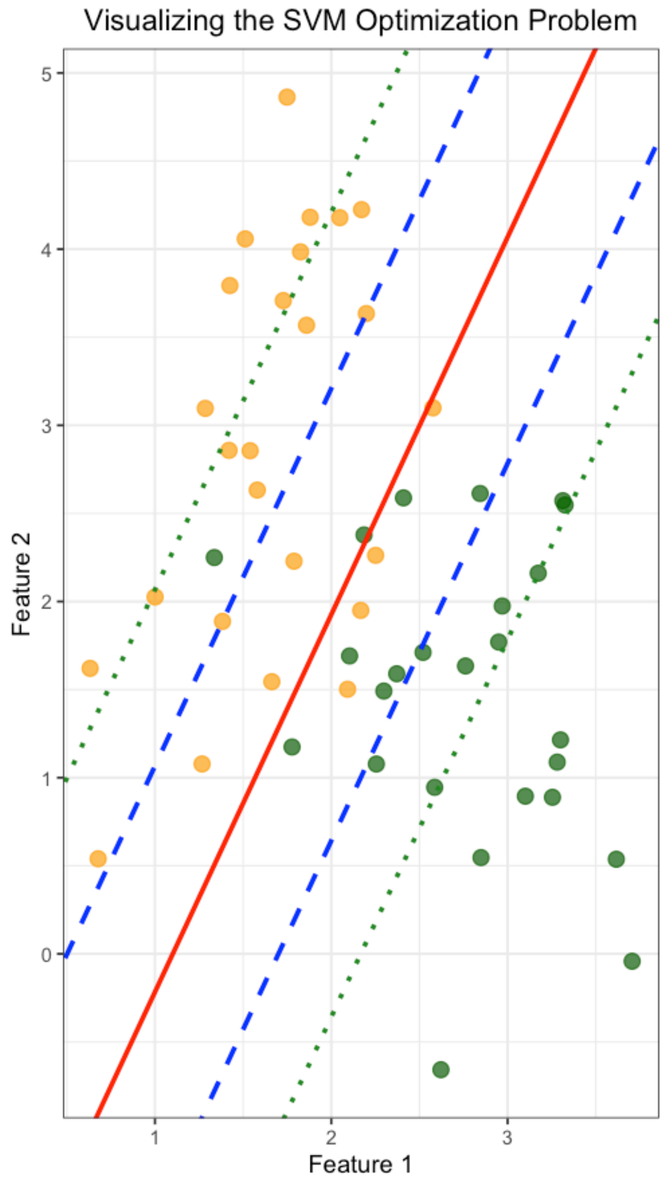


fig3

Large C: high tolerance for observations being on the wrong side

Small C: little tolerance for observations being on the wrong side.

Non-linear SVCs

Enlarging feature space using transformations

Sometimes a linear boundary won't work!

Enlarge the space of features by <u>including</u> transformations;

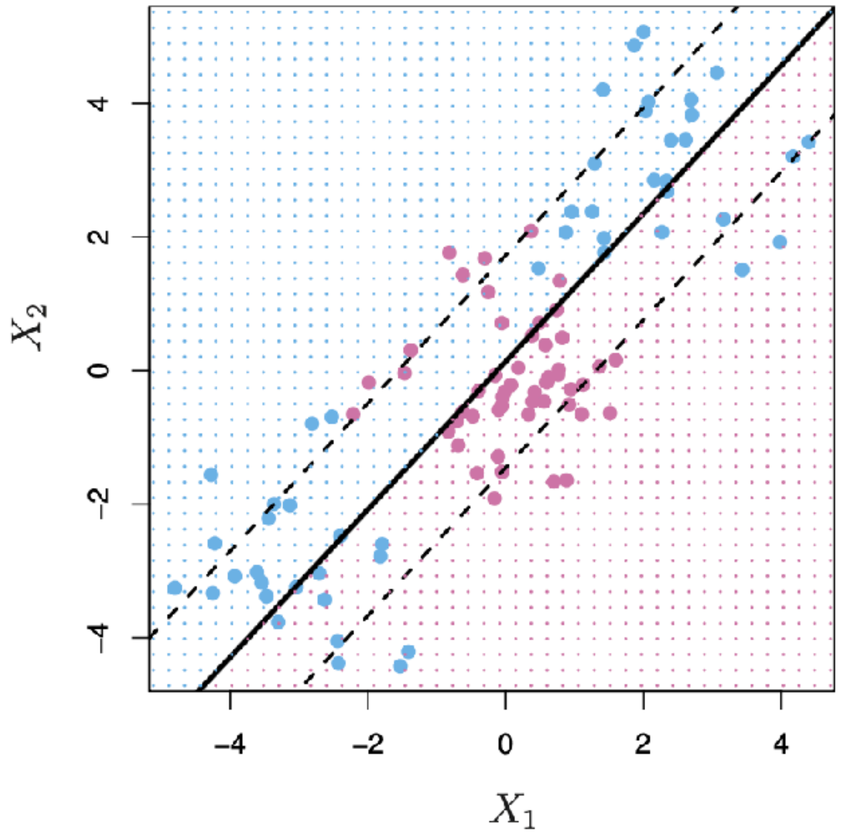
Example: $X_1^2, X_2^2, X_1X_2,$ and $X_1X_2^2$.

Go from a p-dimensional to an $M > p$ dimensional feature space. Then, fit a support-vector classifier in the enlarged space.

Results in non-linear decision boundaries in the original feature space.

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 + \beta_6 X_1 X_2^2 \lessgtr 0$$

Polynomial transformations scale very quickly:
how many parameters for cubic polynomials?



Tackling non-linearities using SVCs

Inner products and kernel functions (🤢)

Inner product of two vectors $\boldsymbol{x}_i$ and $\boldsymbol{y}_i$: $\langle \boldsymbol{x}_i, \boldsymbol{y}_i \rangle = \sum_{j=1}^{p} \boldsymbol{x}_{ij} \boldsymbol{y}_{ij}$

Linear SVC: $f(\boldsymbol{x}) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle \boldsymbol{x}, \boldsymbol{x}_i \rangle$. #parameters = #samples + 1

Support Vector **Machines (SVMs)**: use kernel functions instead of inner products

$$f(\boldsymbol{x}) = \beta_0 + \sum_{i=1}^{n} \alpha_i \, K\left(\boldsymbol{x}, \boldsymbol{x}_i\right)$$

$$K\left(\boldsymbol{y}, \boldsymbol{x}_i\right) = \left(1 + \sum_{j=1}^{p} y_j x_{ij}\right)^d$$

Polynomial kernel

$$K\left(\boldsymbol{y}, \boldsymbol{x}_i\right) = \exp\left(-\gamma \sum_{j=1}^{p} |y_j - x_{ij}|\right)$$

$\mathscr{L}_1$-norm kernel

$$K\left(\boldsymbol{y}, \boldsymbol{x}_i\right) = \exp\left(-\gamma \sum_{j=1}^{p} \left(y_j - x_{ij}\right)^2\right)$$

Radial kernel

SVMs: more than 2 classes?

Two competing approaches

SVMs as defined work for $K = 2$ classes. What about $K > 2$ classes?

OVA: One-versus-All

1. Fit K different 2-class SVM classifiers:
$$\hat{f}_k(\boldsymbol{x}) \text{ for } k = 1, \ldots, K$$
$\hat{f}_k(\boldsymbol{x}^\star)$ classifies $\boldsymbol{x}^\star$ as class k vs 'other'.

2. Classify $\boldsymbol{x}^\star$ to the class $k^\star$, where
$$k^\star = \text{argmax}_k \hat{f}_k(\boldsymbol{x}^\star)$$

Computationally lighter!

OVO: One-versus-One

1. Fit all $\binom{K}{2}$ pairwise classifiers:
$$\hat{f}_{kl}(\boldsymbol{x}) \text{ for } 1 \leq k < l \leq K$$

2. Classify $\boldsymbol{x}^\star$ to the class $k^\star$ that wins most pairwise comparisons.

Pick this when K is small

Post-class debrief

Today we discussed…

The Core Idea: Find the Best Separating Hyperplane

Hyperplane: A line (in 2D), a plane (in 3D), or a flat subspace that separates classes.
Goal: Find the optimal hyperplane that creates the largest possible margin (gap) between the classes.

- **Maximal Margin Classifier:** Finds the hyperplane with the widest margin. Limitation: Only works for perfectly linearly separable data. The points on the margin's edge are support vectors.
- **Support Vector Classifier (Soft Margin):** Allows some data points to be on the wrong side of the margin (or even the hyperplane). The Cost parameter tunes the trade-off between margin width and the number of violations.
- **Support Vector Machine (Non-Linear):** Uses the kernel trick to create non-linear decision boundaries by implicitly mapping features to a higher-dimensional space where they become separable.

SVMs for multi-class problems

- One-vs-One (OVO): Creates a classifier for every pair of classes.
- One-vs-All (OVA): Creates a classifier for each class against all the others.

That's all Folks!